

A Superfast Direct Solver for Nonuniform Discrete Fourier Transform of Type-III

Jingyu Liu

School of Mathematical Sciences, Fudan University

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Joint work with Prof. Yingzhou Li

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- 4 Numerical Examples

Nonuniform discrete Fourier transform (NUDFT):

$$f_j = \sum_{k=0}^{N-1} c_k e^{-2\pi i \omega_k x_j}, \quad 0 \leq j \leq M-1.$$

- Sample points: $\{x_j\}_{j=0}^{M-1} \subset [0, 1)$.
- Frequencies: $\{\omega_k\}_{k=0}^{N-1} \subset [-1/2, N-1/2)$.
- Coefficients: $\{c_k\}_{k=0}^{N-1} \subset \mathbb{C}$.
- Target values: $\{f_j\}_{j=0}^{M-1} \subset \mathbb{C}$.
- NUDFT matrix: $\mathbf{A} = (e^{-2\pi i x_j \omega_k})_{j,k} \in \mathbb{C}^{M \times N}$.

Forward and Inverse Problems

Forward problem: $\mathbf{f} = \mathbf{A}\mathbf{c}$.

- Uniform case (FFT): $M = N$, $x_j = j/M$ and $\omega_k = k$.
Complexity: $\mathcal{O}(N \log N)$ [Cooley and Tukey 1965].
- Nonuniform case (NUFFT)
 - Type-I: Equispaced sample points, i.e., $x_j = j/M$;
 - Type-II: Continuous integer frequencies, i.e., $\omega_k = k$;
 - Type-III: Both sample points and frequencies are nonuniform.

Complexity: $\mathcal{O}(M + N + N \log N)$ [Barnett, Magland, and Klinteberg 2019].

Inverse problem: $\mathbf{c} = \mathbf{A}^\dagger \mathbf{f}$.

- Iterative methods: CG + NUFFT.
- Direct methods: [Kircheis and Potts 2019], [Wilber, Epperly, and Barnett 2025].

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Displacement Structure of NUDFT-II Matrix

Method in [Wilber, Epperly, and Barnett 2025]:

$\mathbf{A} = (e^{-i2\pi\omega_k x_j})$ is a Vondermonde matrix satisfying the following Sylvester equation:

$$\mathbf{\Gamma A} - \mathbf{A C} = \mathbf{u e}_{N-1}^\top.$$

- $\mathbf{\Gamma} = \text{diag}(\gamma_0, \gamma_1, \dots, \gamma_{M-1})$ with $\gamma_j = e^{-2\pi i x_j}$;
- $\mathbf{C} = \begin{bmatrix} \mathbf{0} & \mathbf{1} \\ \mathbf{I}_{N-1} & \mathbf{0} \end{bmatrix}$ is a circulant shift matrix;
- $u_j = \gamma_j^N - 1$;
- $\mathbf{e}_{N-1} = [0 \ \cdots \ 0 \ 1]^\top$.

Displacement Structure of NUDFT-II Matrix

Sylvester equation:

$$\Gamma \mathbf{A} - \mathbf{A} \mathbf{C} = \mathbf{u} \mathbf{e}_{N-1}^{\top}.$$

Fact: $\mathbf{C}$ can be diagonalized using the DFT matrix, i.e., $\mathbf{C} = \mathbf{F}^{-1} \mathbf{D} \mathbf{F}$

- $\mathbf{F} = (e^{-2\pi i j k / N})_{j,k};$
- $\mathbf{D} = \text{diag}(\xi_0, \xi_1, \dots, \xi_{N-1})$ with $\xi_k = e^{-2\pi i k / N}.$

Replacing $\mathbf{C}$ with $\mathbf{F}^{-1} \mathbf{D} \mathbf{F}$ in the Sylvester equation, we have

$$\begin{aligned} \Gamma \mathbf{A} - \mathbf{A} \mathbf{F}^{-1} \mathbf{D} \mathbf{F} &= \mathbf{u} \mathbf{e}_{N-1}^{\top} \\ \implies \Gamma (\mathbf{A} \mathbf{F}^{-1}) - (\mathbf{A} \mathbf{F}^{-1}) \mathbf{D} &= \mathbf{u} (\mathbf{e}_{N-1}^{\top} \mathbf{F}^{-1}). \end{aligned}$$

Displacement Structure of NUFT-II Matrix

Let $\tilde{\mathbf{A}} = \mathbf{A}\mathbf{F}^{-1}$ then

$$\mathbf{\Gamma}\tilde{\mathbf{A}} - \tilde{\mathbf{A}}\mathbf{D} = \mathbf{u}\mathbf{v}^\top.$$

- $\mathbf{\Gamma} = \text{diag}(\gamma_0, \gamma_1, \dots, \gamma_{M-1})$ with $\gamma_j = e^{-2\pi i x_j}$;
- $\mathbf{D} = \text{diag}(\xi_0, \xi_1, \dots, \xi_{N-1})$ with $\xi_k = e^{-2\pi i k/N}$;
- $u_j = \gamma_j^N - 1$;
- $v_k = \xi_k/N$.

Displacement Structure of NUDFT-II Matrix

$\tilde{\mathbf{A}}$ has a **displacement structure** with displacement rank 1

$$\mathbf{\Gamma} \tilde{\mathbf{A}} - \tilde{\mathbf{A}} \mathbf{D} = \mathbf{u} \mathbf{v}^\top.$$

For a row index set $\mathcal{J}$ and a column index set $\mathcal{K}$, the submatrix $\tilde{\mathbf{A}}_{\mathcal{J},\mathcal{K}}$ also satisfies the Sylvester equation:

$$\mathbf{\Gamma}_{\mathcal{J}} \tilde{\mathbf{A}}_{\mathcal{J},\mathcal{K}} - \tilde{\mathbf{A}}_{\mathcal{J},\mathcal{K}} \mathbf{D}_{\mathcal{K}} = \mathbf{u}_{\mathcal{J}} \mathbf{v}_{\mathcal{K}}^\top.$$

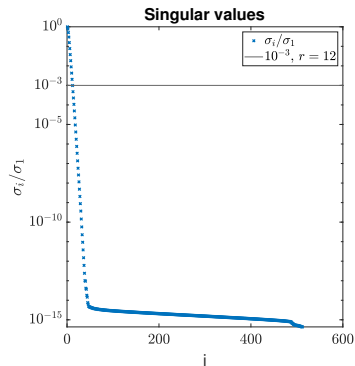
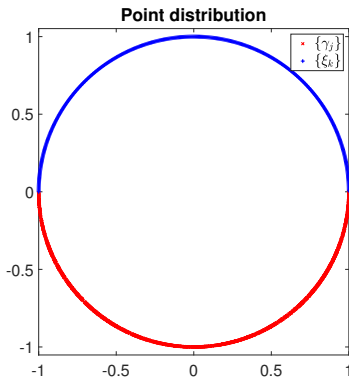
When $\{\gamma_j\}_{j \in \mathcal{J}}$ and $\{\xi_k\}_{k \in \mathcal{K}}$ are **disjoint**:

- ~> The singular values of $\tilde{\mathbf{A}}_{\mathcal{J},\mathcal{K}}$ **decay exponentially** [Beckermann and Townsend 2017];
- ~> **Low-rank property:** $\tilde{\mathbf{A}}_{\mathcal{J},\mathcal{K}}$ can be approximated by a **low-rank** matrix.

Low-Rank Property in the NUDFT-II Matrix

$\tilde{\mathbf{A}}$ is a Cauchy-like matrix:

$$\tilde{\mathbf{A}}(j, k) = \frac{u_j v_k}{\gamma_j - \xi_k}.$$



HSS Matrix

Idea: Approximate $\tilde{\mathbf{A}}$ by a hierarchical semiseparable (HSS) matrix.

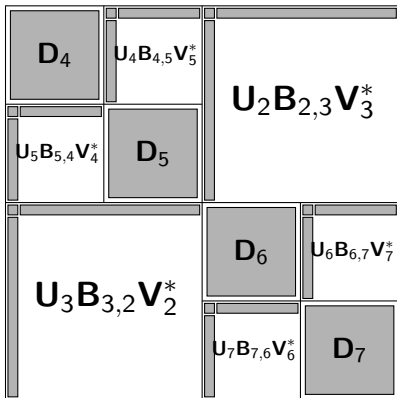


$$\mathbf{D}_1 = \begin{bmatrix} \mathbf{D}_2 & \mathbf{U}_2 \mathbf{B}_{2,3} \mathbf{V}_3^* \\ \mathbf{U}_3 \mathbf{B}_{3,2} \mathbf{V}_2^* & \mathbf{D}_3 \end{bmatrix},$$

recursion on the diagonal: $\mathbf{D}_2 = \begin{bmatrix} \mathbf{D}_4 & \mathbf{U}_4 \mathbf{B}_{4,5} \mathbf{V}_5^* \\ \mathbf{U}_5 \mathbf{B}_{5,4} \mathbf{V}_4^* & \mathbf{D}_6 \end{bmatrix},$

recursion on the off-diagonal: $\mathbf{U}_2 = \begin{bmatrix} \mathbf{U}_4 \mathbf{R}_4 \\ \mathbf{U}_5 \mathbf{R}_5 \end{bmatrix}$ and $\mathbf{V}_2 = \begin{bmatrix} \mathbf{V}_4 \mathbf{W}_4 \\ \mathbf{V}_5 \mathbf{W}_5 \end{bmatrix}.$

HSS Matrix



$$\mathbf{U}_2 = \begin{bmatrix} \mathbf{U}_4 \mathbf{R}_4 \\ \mathbf{U}_5 \mathbf{R}_5 \end{bmatrix}, \quad \mathbf{V}_2 = \begin{bmatrix} \mathbf{V}_4 \mathbf{W}_4 \\ \mathbf{V}_5 \mathbf{W}_5 \end{bmatrix}.$$

$$\mathbf{U}_3 = \begin{bmatrix} \mathbf{U}_6 \mathbf{R}_7 \\ \mathbf{U}_6 \mathbf{R}_7 \end{bmatrix}, \quad \mathbf{V}_3 = \begin{bmatrix} \mathbf{V}_6 \mathbf{W}_6 \\ \mathbf{V}_7 \mathbf{W}_7 \end{bmatrix}.$$

Shared basis property: $\mathbf{U}_\tau$ is the column basis of $\tilde{\mathbf{A}}_{\text{HSS}}(\mathcal{J}_\tau, \mathcal{K}_\tau^c)$:

$$\tilde{\mathbf{A}}_{\text{HSS}}(\mathcal{J}_4, :) = \begin{bmatrix} \mathbf{D}_4 & \mathbf{U}_4 \mathbf{B}_{4,5} \mathbf{V}_5^* & \mathbf{U}_4 \mathbf{R}_4 \mathbf{B}_{2,3} \mathbf{W}_6^* \mathbf{V}_6^* & \mathbf{U}_4 \mathbf{R}_4 \mathbf{B}_{2,3} \mathbf{W}_7^* \mathbf{V}_7^* \end{bmatrix}$$

A Superfast Direct Solver for Type-II NUDFT Problem

Algorithm A direct solver for type-II INUDFT problem.

- 1: Approximate $\tilde{\mathbf{A}} = \mathbf{A}\mathbf{F}^{-1}$ by an HSS matrix $\tilde{\mathbf{A}}_{\text{HSS}}$.
 - 2: Solve $\mathbf{v} = \arg \min_{\mathbf{w}} \|\tilde{\mathbf{A}}_{\text{HSS}}\mathbf{w} - \mathbf{f}\|_2$ by the HSS least-squares solver.
 - 3: Compute $\mathbf{c} = \mathbf{F}^{-1}\mathbf{v}$ using the iFFT.
-

Type-II approximation: $\mathbf{A} = \tilde{\mathbf{A}}\mathbf{F} \approx \tilde{\mathbf{A}}_{\text{HSS}}\mathbf{F} =: \mathbf{A}_{\text{fast}}$.

HSS rank: $r = \mathcal{O}(\log(1/\varepsilon) \log(N))$ [Wilber, Epperly, and Barnett 2025].

Construction: Factorized alternating direction implicit (fADI) [Benner, Li, and Truhar 2009].

Complexity: $\mathcal{O}((M+N)r^2)$.

HSS least-squares solver: URV factorization [Xi et al. 2014].

Complexity: $\mathcal{O}((M+N)r^2)$ for the factorization and $\mathcal{O}((M+N)r)$ for the each solution.

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Type-II approximation: $\mathbf{A} = \tilde{\mathbf{A}}\mathbf{F} \approx \tilde{\mathbf{A}}_{\text{HSS}}\mathbf{F} =: \mathbf{A}_{\text{fast}}$.

Complexities:

- **Building the solver:** $\mathcal{O}((M+N)r^2) = \mathcal{O}((M+N)\log^2(N))$.
- **Each solution:** $\mathcal{O}((M+N)r + N\log(N)) = \mathcal{O}((M+N)\log(N))$.

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Idea: Basis expansion.

$$\mathbf{A}(j, k) = e^{-2\pi i x_j \omega_k} = \underbrace{\sum_{\ell \in \mathcal{I}_{N;\sigma}} e^{-2\pi i x_j \ell} Q_{\ell,k}}_{\text{Approximation}} + \underbrace{\sum_{\ell \notin \mathcal{I}_{N;\sigma}} e^{-2\pi i x_j \ell} Q_{\ell,k}}_{\text{Error}}.$$

- $\mathcal{I}_{N;\sigma} = \{\ell \in \mathbb{Z} : -(\sigma N - N)/2 \leq \ell \leq (\sigma N + N)/2 - 1\}$: σ -expanded set;
- $Q_{\ell,k} = G(\ell - \omega_k)$ where $G(\omega) = e^{\pi i \omega \frac{\sin(\pi \omega)}{\pi \omega}}$: kernel matrix.

Matrix form:

$$\mathbf{A} = \mathbf{BQ} + \mathbf{E}.$$

Matrix form:

$$\mathbf{A} = \mathbf{B}\mathbf{Q} + \mathbf{E}.$$

Proposition (Li and L., 2025)

If $\{x_j\}$ are i.i.d. uniform random variables in $[0, 1)$ and $\{\omega_k = k + \delta_k\}$ are perturbations of the continuous integers with $|\delta_k| \leq \alpha$ for some constant $\alpha \in [0, 1/2)$. Then $\mathbb{E}\mathbf{E} = \mathbf{0}$ and

$$\mathbb{E}\|\mathbf{E}\|_{\text{F}}^2 \leq \frac{2 \sin^2(\pi\alpha)}{\pi^2((\sigma-1)N/2+1/2-\alpha)} \|\mathbf{A}\|_{\text{F}}^2.$$

Denoting $\mathbf{H} = \mathbf{B}^\dagger \mathbf{A}$, we have

$$\begin{aligned}\mathbf{H} &= \mathbf{Q} + \mathbf{B}^\dagger \mathbf{E}, \\ \mathbf{A} &= \mathbf{B}\mathbf{H} + (\mathbf{I} - \mathbf{B}\mathbf{B}^\dagger)\mathbf{E}.\end{aligned}$$

- $\mathbf{B}$ is a **NUDFT-II** matrix;
- $\mathbf{Q}$ is a kernel matrix, can be compressed into an HSS matrix;
- $\mathbf{H}$ can be empirically approximated by an **HSS matrix**;
- $\mathbf{I} - \mathbf{B}\mathbf{B}^\dagger$ is a projection matrix.

A Superfast Direct Solver for Type-III INUDFT Problem

Algorithm A direct solver for type-III INUDFT problem.

- 1: Construct a type-II direct solver $\mathbf{B}_{\text{fast}} \approx \mathbf{B}$.
 - 2: Compress $\mathbf{B}_{\text{fast}}^\dagger \mathbf{A}$ into an HSS matrix $\mathbf{H}_{\text{HSS}}$.
 - 3: Compute $\mathbf{c} = \mathbf{H}_{\text{HSS}}^{-1} \mathbf{B}_{\text{fast}}^\dagger \mathbf{f}$.
-

Type-III approximation: $\mathbf{A} \approx \mathbf{B}_{\text{fast}} \mathbf{H}_{\text{HSS}}$.

Goal: Find an efficient way to construct an HSS compression of $\mathbf{B}_{\text{fast}}^\dagger \mathbf{A}$.

- Fast computation on $\mathbf{A}$: NUFFT.
- Fast computation on $\mathbf{B}_{\text{fast}}^\dagger$: URV solution and iFFT.

Black-box setting: Only fast matrix-vector multiplication is available.

$\leadsto$ [Levitt and Martinsson 2024]: Using **random samplings**.

Complexity: $\mathcal{O}(Nr^2 + T_{\text{mult}}r)$, where T_{mult} is the complexity of apply $\mathbf{B}_{\text{fast}}^\dagger \mathbf{A}$ to a vector.
Empirically, $r = \mathcal{O}(\log(N))$.

A Superfast Direct Solver for Type-III INUDFT Problem

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 - 3: Compute $\mathbf{c} = \mathbf{H}_{\text{HSS}}^{-1} \mathbf{B}_{\text{fast}}^\dagger \mathbf{f}$.
-

Type-III approximation: $\mathbf{A} \approx \mathbf{B}_{\text{fast}} \mathbf{H}_{\text{HSS}}$.

Complexities:

- **Preliminary:** Construct and factorize $\mathbf{B}_{\text{fast}} \rightsquigarrow \mathcal{O}((M + N) \log^2(N))$.
- **Construction:** Compress $\mathbf{H}_{\text{HSS}} \rightsquigarrow \mathcal{O}((M + N) \log^2(N))$.
- **Factorization:** Factorize $\mathbf{H}_{\text{HSS}} \rightsquigarrow \mathcal{O}(N \log^2(N))$.
- **Each solution:** $\mathbf{c} = \mathbf{H}_{\text{HSS}}^{-1} \mathbf{B}_{\text{fast}}^\dagger \mathbf{f} \rightarrow \mathcal{O}((M + N) \log(N))$.

A Superfast Direct Solver for Type-III INUDFT Problem

Algorithm A direct solver for type-III INUDFT problem.

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 - 3: Compute $\mathbf{c} = \mathbf{H}_{\text{HSS}}^{-1} \mathbf{B}_{\text{fast}}^\dagger \mathbf{f}$.
-

Type-III approximation: $\mathbf{A} \approx \mathbf{B}_{\text{fast}} \mathbf{H}_{\text{HSS}}$.

Complexities:

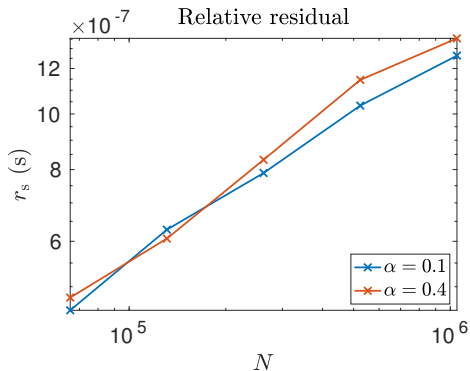
- **Building the solver:** $\mathcal{O}((M + N) \log^2(N))$.
- **Each solution:** $\mathcal{O}((M + N) \log(N))$.

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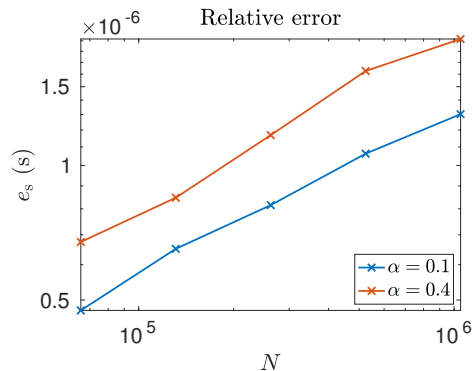
Experiment Setting

- $M = 4N$ and $N = 2^{17}, 2^{18}, 2^{19}, 2^{20}$.
- Samples are i.i.d. uniform random variables in $[0, 1)$.
- Frequencies are perturbed integers given by $\omega_k = k + \alpha\psi_k$ where $\{\psi_k\}$ are i.i.d. uniform random variables on $[-1, 1]$ and $\alpha \in \{0.1, 0.4\}$.

Direct Solver



(a) Relative residual.



(b) Relative error.

N	method	t_{pre}	t_{iter}	n_{iter}	r_s	e_s
2^{17}	lsqr	-	$4.3\text{e}+01$	500	$5.9\text{e}-06$	$1.7\text{e}-03$
	plsqr	$2.3\text{e}+02$	$1.4\text{e}+01$	5	$2.8\text{e}-13$	$1.6\text{e}-11$
2^{18}	lsqr	-	$8.4\text{e}+01$	500	$2.7\text{e}-07$	$1.1\text{e}-05$
	plsqr	$5.2\text{e}+02$	$2.9\text{e}+01$	5	$7.8\text{e}-14$	$4.9\text{e}-13$
2^{19}	lsqr	-	$1.7\text{e}+02$	500	$9.6\text{e}-09$	$3.0\text{e}-07$
	plsqr	$1.2\text{e}+03$	$6.2\text{e}+01$	5	$3.2\text{e}-13$	$1.1\text{e}-12$
2^{20}	lsqr	-	$5.2\text{e}+02$	500	$9.7\text{e}-07$	$6.2\text{e}-05$
	plsqr	$2.6\text{e}+03$	$1.7\text{e}+02$	5	$1.2\text{e}-13$	$7.6\text{e}-13$

Table: Time and error results of the iterative solver under the random-perturbation case, $\alpha = 0.1$.

N	method	t_{pre}	t_{iter}	n_{iter}	r_s	e_s
2^{17}	lsqr	-	$4.3\text{e}+01$	500	$5.1\text{e}-10$	$9.1\text{e}-09$
	plsqr	$2.3\text{e}+02$	$2.6\text{e}+01$	9	$8.2\text{e}-15$	$3.2\text{e}-14$
2^{18}	lsqr	-	$8.5\text{e}+01$	500	$1.2\text{e}-07$	$6.3\text{e}-06$
	plsqr	$5.1\text{e}+02$	$5.2\text{e}+01$	9	$9.0\text{e}-14$	$5.6\text{e}-13$
2^{19}	lsqr	-	$2.2\text{e}+02$	500	$2.1\text{e}-06$	$6.4\text{e}-04$
	plsqr	$1.1\text{e}+03$	$9.4\text{e}+01$	8	$2.4\text{e}-13$	$2.6\text{e}-11$
2^{20}	lsqr	-	$4.8\text{e}+02$	500	$4.3\text{e}-06$	$5.5\text{e}-04$
	plsqr	$2.6\text{e}+03$	$2.4\text{e}+02$	9	$7.1\text{e}-14$	$1.9\text{e}-12$

Table: Time and error results of the iterative solver under the random-perturbation case, $\alpha = 0.4$.

Conclusions:

- A superfast direct solver for type-III NUFT problem;
- The direct solver could also be used as an efficient preconditioner;
- Error bound of the forward approximation.

Future work:

- Extension to high-dimensional problems;
- Method for regularized least-squares problems.